

Workshop: Differentiation Chain Rule

UEA LEARNING ENHANCEMENT TEAM MATHEMATICS & STATISTICS

Find the derivative of:

1. $y = (5x + 4)^5$

$$u = 5x + 4$$

$$y = u^5$$

$$\frac{du}{dx} = 5$$

$$\frac{dy}{du} = 5u^4$$

$$\frac{dy}{dx} = 5u^4 \times 5 = 25(5x + 4)^4$$

2. $y = (5x^3 - x - 1)^9$

$$\frac{dy}{dx} = 9(15x^2 - 1)(5x^3 - x - 1)^8$$

3. $y = (4x + 9)^{-1}$

$$\frac{dy}{dx} = -1(4x + 9)^{-2} \times 4 = -4(4x + 9)^{-2}$$

4. $y = \sqrt{(8x + 1)}$

$$y = (8x + 1)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(8x + 1)^{-\frac{1}{2}} \times 8 = \frac{4}{\sqrt{8x + 1}}$$

5. $y = (e^x - 2x)^4$

$$\frac{dy}{dx} = 4(e^x - 2x)^3(e^x - 2)$$

Find the derivative of:

6. $y = e^{4x}$

$$u = 4x$$

$$y = e^u$$

$$\frac{du}{dx} = 4$$

$$\frac{dy}{du} = e^u$$

$$\frac{dy}{dx} = 4e^u = 4e^{4x}$$

7. $y = 5e^{x^2}$

$$\frac{dy}{dx} = 5e^{x^2} \times 2x = 10xe^{x^2}$$

8. $y = 2e^{x^3+2x-4}$

$$\frac{dy}{dx} = 2(3x^2 + 2)e^{x^3+2x-4}$$

9. $y = e^{4-x-x^2}$

$$\frac{dy}{dx} = (-1 - 2x)e^{4-x-x^2}$$

10. $y = 6e^{1+\frac{1}{x}}$

$$y = 6e^{1+x^{-1}}$$

$$\frac{dy}{dx} = 6e^{1+\frac{1}{x}} \times (-x^{-2}) = -\frac{6e^{1+\frac{1}{x}}}{x^2}$$

Mixed

Use the examples above and your understanding of the chain rule to differentiate:

11. $y = (8x + 4)^{-4}$,

$$\frac{dy}{dx} = -4(8x + 4)^{-5} \times 8 = -32(8x + 4)^{-5}$$

$$12. y = e^{1+\ln x}$$

$$\frac{dy}{dx} = \frac{1}{x} e^{1+\ln x}$$

$$13. y = \ln(x^2 + 3)$$

$$\frac{dy}{dx} = \frac{1}{x^2 + 3} \times 2x = \frac{2x}{x^2 + 3}$$

$$14. y = 5t + e^{6t-1}$$

$$\frac{dy}{dt} = 5 + 6e^{6t-1}$$

$$15. S = 4 \ln(5x^2 + 2x^4) + (x - 1)^5$$

$$\begin{aligned} \frac{dS}{dx} &= 4(10x + 8x^3) \times \frac{1}{5x^2 + 2x^4} + 5(x - 1)^4 \\ &= \frac{8x(5 + 4x^2)}{5x^2 + 2x^4} + 5(x - 1)^4 \end{aligned}$$

$$16. P = e^{(3t+5)^6}$$

$$u = (3t + 5)^6$$

$$P = e^u$$

$$\frac{du}{dt} = 3 \times 6(3t + 5)^5 = 18(3t + 5)^5$$

$$\frac{dP}{du} = e^u$$

$$\frac{dP}{dt} = 18(3t + 5)^5 e^{(3t+5)^6}$$